

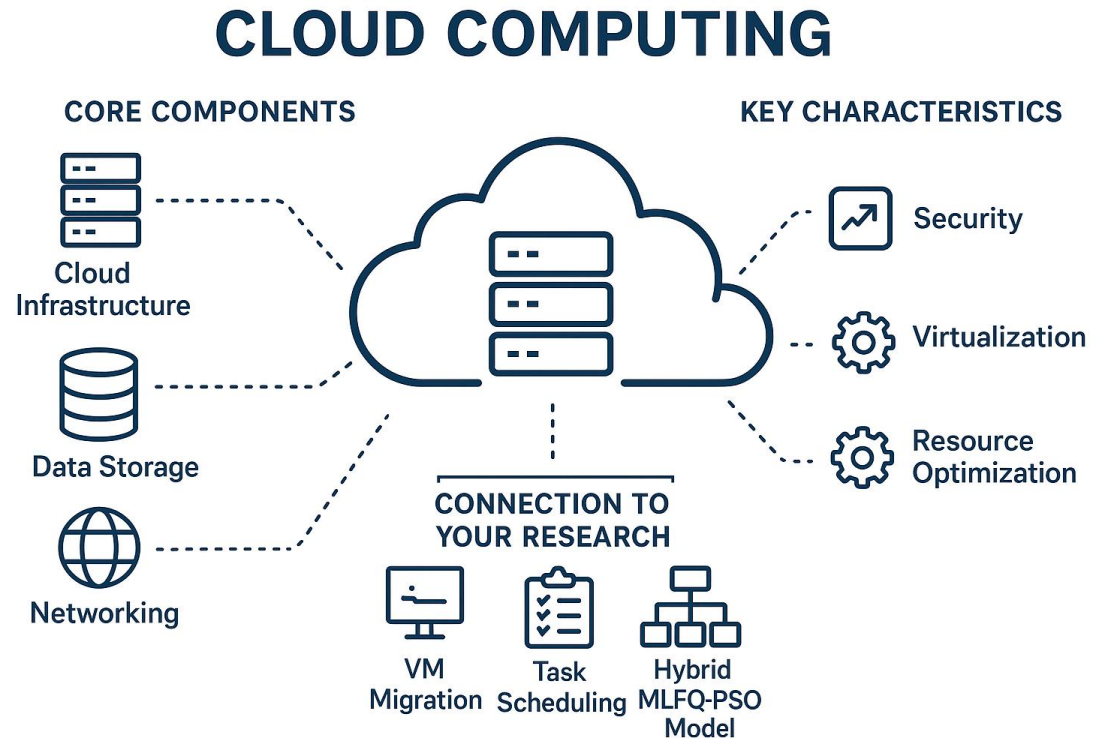
Cloud Infrastructure for AI (AI-601)

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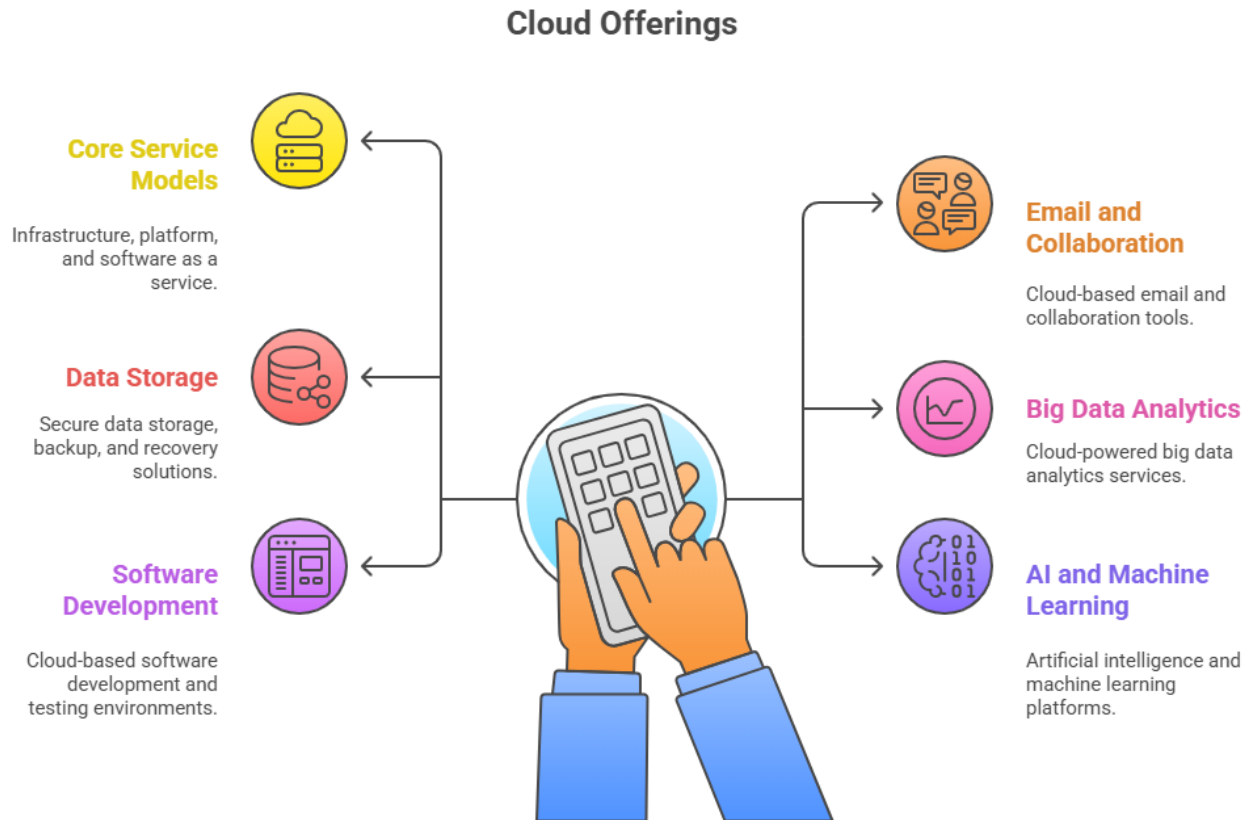


What Is Cloud Computing?

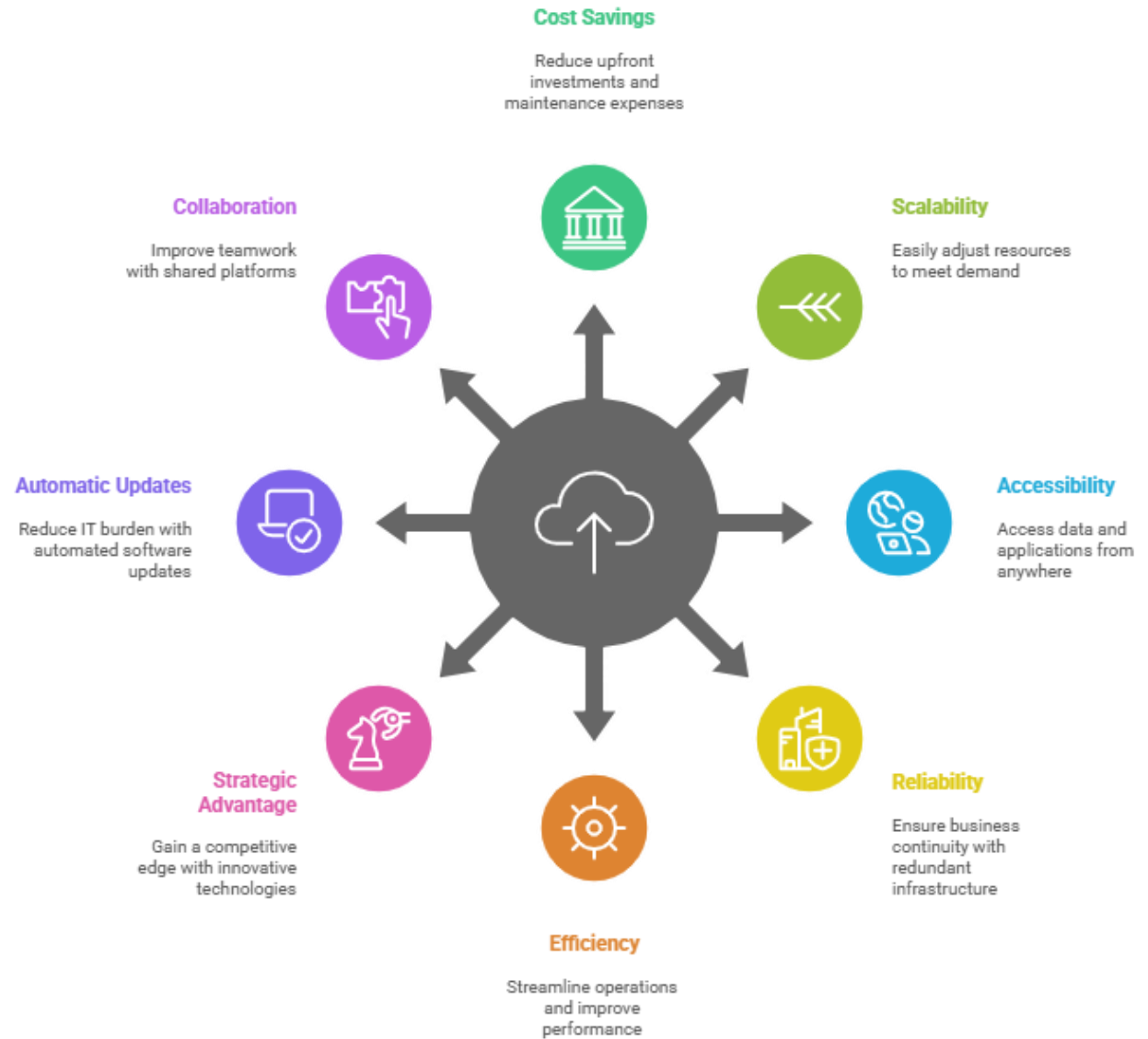


According to NIST, Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

Offerings of Cloud



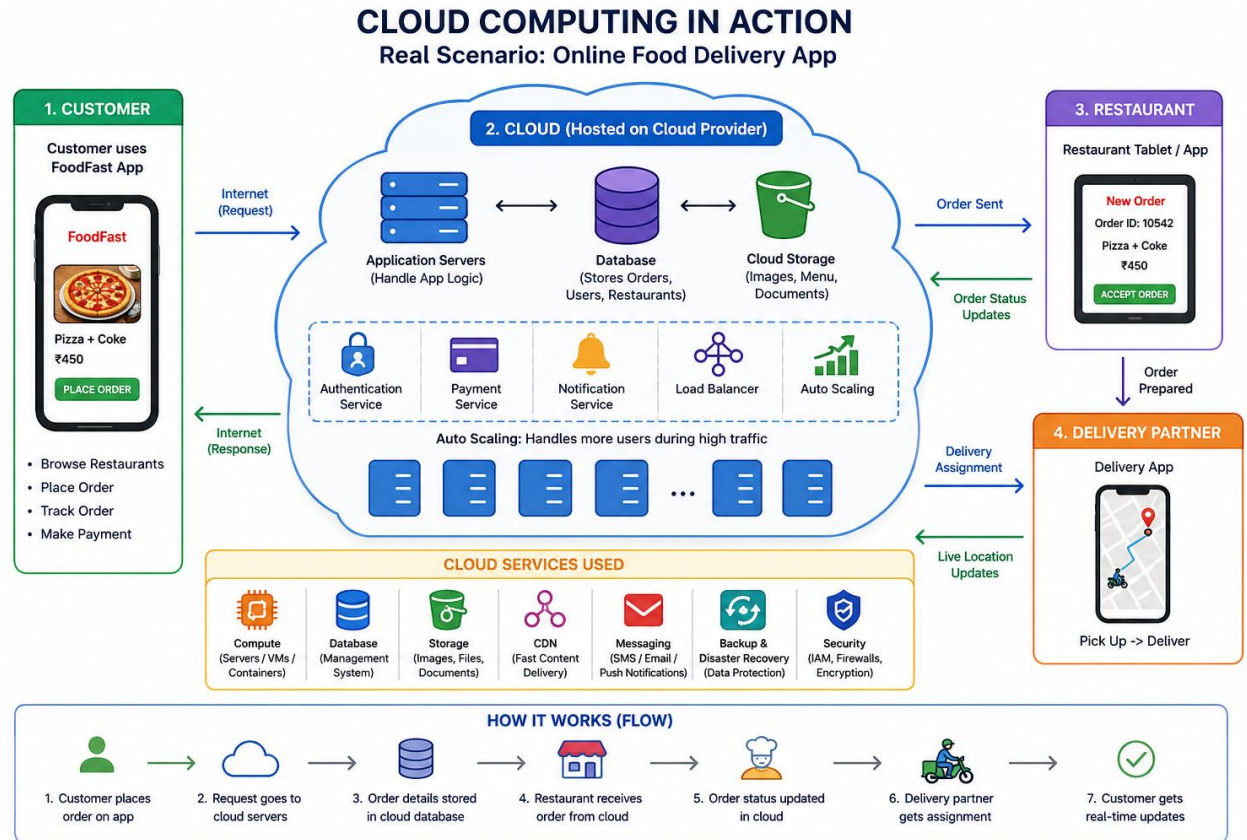
Goals of Cloud Computing



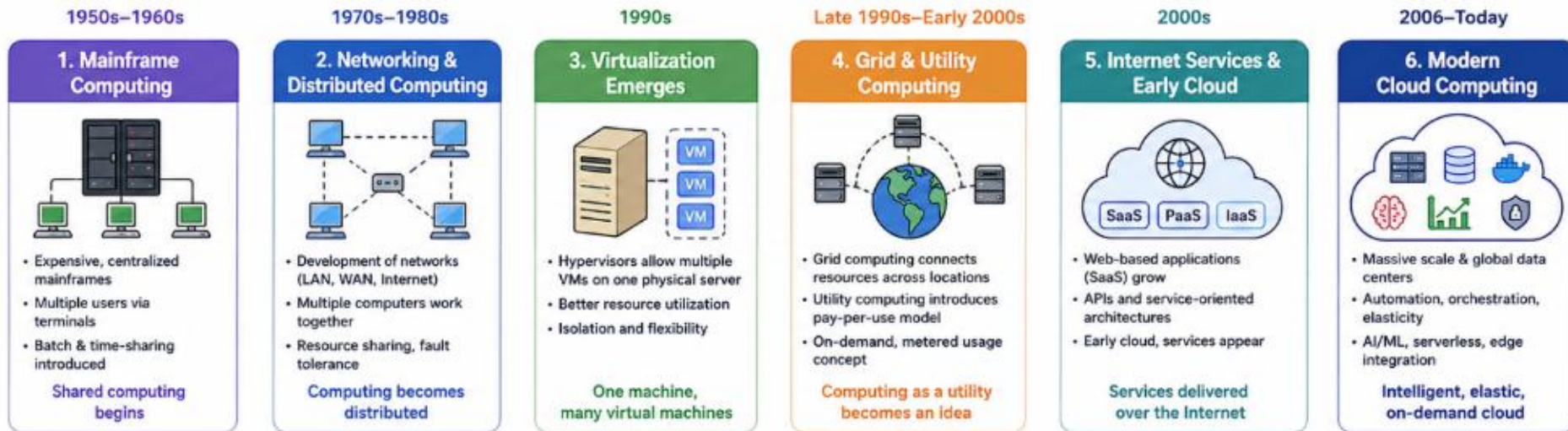
Benefits of Cloud

- Cloud computing do not need high quality equipment for user, and it is very easy to use.
- Provides dependable and secure data storage center.
- Reduce run time and response time.
- Cloud is a large resource pool that you can buy on-demand service.
- Scale of cloud can extend dynamically providing nearly infinite possibility for users to use internet.

How Cloud Computing works



Evolution of Cloud Computing



Evolution of Cloud Computing

1950s–1960s



Mainframe Computing

- Early computers were large, expensive, and shared by many users.
- The mainframe divided its processing time among multiple users.
- Shared computing resources, multiple users, centralized infrastructure, and computing as a shared resource.

1960s–1980s



Networking & Distributed Computing

- Multiple computers could cooperate instead of one doing all the work.
- Introduced resource sharing, communication between computers, distributed data, fault tolerance, and workload distribution.

1990s



Virtualization & Internet Growth

- The Internet made it possible to access computing services remotely.
- A physical server could be divided into multiple virtual machines.
- Allowed providers to share physical hardware, isolate users, increase utilization, create servers quickly, and scale easily.

1990s–2000s



Grid & Utility Computing

- Organizations combined computing resources from multiple locations for large computational problems.
- Computing viewed like a utility where users pay for what they consume.
- Introduced metered usage, on-demand resources, and service-based computing.

1999–2006



Cloud Services Emerge

- Software as a Service delivered software through the Internet instead of local installation.
- Salesforce launched its web-based CRM model in 1999.
- Amazon Web Services introduced infrastructure services like Amazon S3 and Amazon EC2 in 2006.
- Established the model of renting compute, storage, and networking over the Internet.

2008–Today



Modern Cloud Computing

- Cloud evolved from renting virtual servers to providing a huge range of managed services.
- Provides infrastructure like virtual machines, storage, and networking.
- Provides platforms like databases, containers, and application platforms.
- Provides advanced services like AI/ML, big-data analytics, serverless computing, IoT, and security.

Computing Paradigms Related to Cloud

Cloud computing did not develop in isolation. It builds upon several earlier computing paradigms.

Paradigm	Main Idea
Ubiquitous Computing	Computing is available everywhere and embedded in everyday devices
Distributed Computing	Multiple computers work together over a network
Parallel Computing	Multiple processors perform tasks simultaneously
Grid Computing	Resources from multiple organizations/sites are coordinated
Utility Computing	Computing is delivered like a utility; pay for what you use
Edge Computing	Processing is moved closer to the users/devices
Cloud Computing	Combines these ideas to provide scalable computing as a service

Cloud vs. Other Computing Paradigms

Feature	Cloud	Distributed	Parallel	Grid	Utility	Ubiquitous
Main focus	Services	Resource sharing	Speed	Coordination	Pay-per-use	Everywhere
Network-based	Yes	Yes	Not necessarily	Yes	Usually	Usually
Scalability	High	High	Moderate/High	High	High	High
Pay-per-use	Yes	Not necessarily	No	Usually no	Yes	No
Processing location	Cloud/DC	Multiple systems	Multiple processors	Multiple sites	Provider	Everywhere

Cloud and Other Computing Paradigms

Ubiquitous computing → Computing everywhere

Distributed computing → Many computers working together

Parallel computing → Many tasks processed simultaneously

Grid computing → Coordinated resources across locations

Utility computing → Pay for what you use

Edge computing → Processing near the user

Cloud computing → Scalable computing delivered as a service